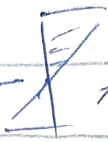
PF Lab - Assignments:-

Activity-2 :-

i-  4min -  7min hourglass.

to Exactly measure 9 minutes. First use the 7-min hourglass. Once its Complete means 7 mins are passed away now. use 4 min hourglass and exactly when the glass has dropped half sand it means 2 minutes have been passed away. So, by this, the time can be measured upto 9-mins.

ii- The final state for the bulb no. 64 is that it should be on. As the perfect Squares has odd no. of divisors this means the Bulbs coming on the no. of perfect Squares will remain illuminated. The final Bulbs which are illuminated are:- Bulb:- 1, 4, 9, 16, 25, 36, 49, 64, 81, 100 only.

Logic:- They all had odd no. of divisors. So initially Bulbs are off then on then off then on

Qno 04:- Swaping of Frogs:-

Initial positions:-

$L_1, L_2, L_3 \ \underline{\qquad}\ R_1, R_2, R_3$

$L_1, L_2, \underline{\qquad}, L_3, R_1, R_2, R_3$

$L_1, L_2, R_1, L_3, \underline{\qquad}, R_2, R_3$

$L_1, L_2, R_1 \ \underline{\qquad}, L_3, R_2, R_3$

$L_1, R_1, L_2, L_3, R_2, R_3$

$L_1, L_1, R_1 \ \underline{\qquad}\ R_3, L_2, L_3$

$R_2, R_2, L_1 \ \underline{\qquad}\ R_3, L_1$

$\downarrow$

$L_1, L_2, L_3 \ \underline{\qquad}\ R_1, R_2, R_3$

$\Rightarrow L_1, L_2, L_3, R_1, \underline{\qquad}, R_2, R_3$

$\Rightarrow L_1, L_2, \underline{\qquad}, R_1, L_3, R_2, R_3$

$\Rightarrow L_1, \underline{\qquad}, L_2, R_1, L_3, R_2, R_3$

$\Rightarrow L_1, R_1, L_2, \underline{\qquad}, L_3, R_2, R_3$

$\Rightarrow L_1, R_1, L_2, R_2, L_3, \underline{\qquad}, R_3$

$\Rightarrow L_1, R_1, L_2, R_2, L_3, R_3, \underline{\qquad}$

$\Rightarrow L_1, R_1, L_2, R_2, \underline{\qquad}, R_3, L_3$

$\Rightarrow L_1, R_1, \underline{\qquad}, R_2, L_2, R_3, L_3$

$\Rightarrow \underline{\qquad}, R_1, L_1, R_2, L_2, R_3, L_3$

$\Rightarrow R_1, \underline{\qquad}, L_1, R_2, L_2, R_3, L_3$

$\Rightarrow R_1, R_2, L_1, \underline{\qquad}, L_2, R_3, L_3$

$\Rightarrow R_1, R_2, L_1, R_3, L_2, \underline{\qquad}, L_3$

$\Rightarrow R_1, R_2, L_1, R_3, \underline{\qquad}, L_2, L_3$

⇒ $R_1, R_2, \_\_\_ R_3, L_1, L_2, L_3$

⇒ $R_1, R_2, R_3 \longrightarrow L_1, L_2, L_3$

Req Condition :-

Activity No 03:-

i- Step I: Input radius (r)
Step II: declare value of
PI = 3.14
Step III: Use formula
PI * r * r.
Step IV: Display Area as
output.

Start → input radius → Area = PI * r * r → output Area → stop.

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ii- Step I: Input diameter
Step II: Calculate radius
By using $d = 2r \Rightarrow r = \frac{d}{2}$
Step III: declare Area_1 = 14.125 inch
Step IV: Calculate Area_2 from
PI * r * r
Step V: Divide Area_2 / Area_1
to calculate the number of slices
in each Pizza.

Start → input d → r = d/2 → Area_1 = 14.125 → Area_2 = PI * r * r → Area_2 / Area_1 → result = A_2 - A_1 → output result → Stop

Activity 3:-

Algo:- Step I:- Input no. of weekly hours(n) of work

Step II:- Check if hours(n) $<= 35$.

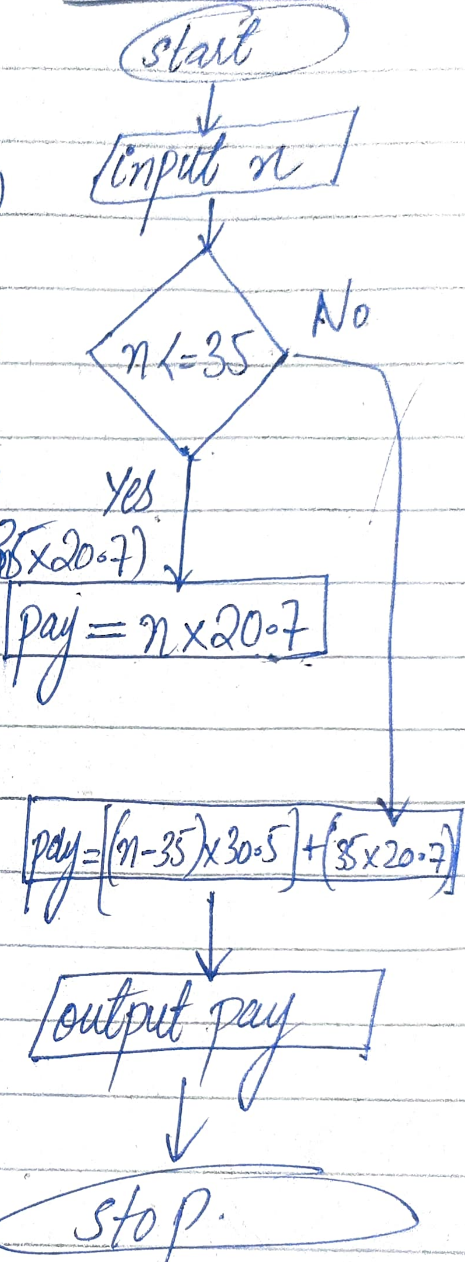
Step III: if $n <= 35$ use $pay = n \times 20.7$

Step IV: else $n > 35$ use $pay = [(n+35) \times 30.5] + (35 \times 20.7)$

Step V: Stop the processing

Step VI: output the Total weekly pay.

Flow-Chart



Qno 04:-

Cost price is required to calculte the selling price of marked Item.

$\because$ Marked price $= 80\%$ of Cp

$\boxed{= 0.8 \times Cp}$

New $Cp = Cp + 0.8\,Cp$

Selling price $= 1.8 \times Cp \times 0.9$

$\boxed{\text{Selling price} = Cp \times 1.62}$

5- Algorithm:-

Step I: Input No of pages (n)

Step II: Calculate the amount for 10 pges and store in variable

Pay = $3.00 + (0.20$ x 10)

Step III: Now in condition check wether the pages (n<=10):- then use

pay_1 = $ 3.00 + (n x 0.20)

else n>10:- then use:-

pay_1 = pay + [(-n+10) x 0.10]

Step IV:- Display pay_1 as output

Flow Chart